

Playing with Numbers — Solution

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main() {
3     int n;
4     scanf("%d", &n);
5     if (n == 0) {
6         printf("1");
7         return 0;
8     }
9     long long a = 1; // ways[0]
10    long long b = 1; // ways[1]
11    long long c = 1; // ways[2]
12
13    for (int i = 3; i <= n; i++) {
14        long long next = c + a;
15        a = b;
16        b = c;
17        c = next;
18    }
19    printf("%lld", c);
20
21    return 0;
22 }
23
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.